Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/29/2021
PI Name:	McLean, Robert Ph.D.		
Project Title:	Polymicrobial Biofilm Growth and Control during Spaceflight		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	San Marcos	**State:**	TX
Zip Code:	78666-4684	**Congressional District:**	35
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/05/2016	**End Date:**	12/04/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/4/2021 per NSSC information (Ed., 3/10/21)		
Key Personnel Changes/Previous PI:	October 2021 report: Key personnel on the original grant application are unchanged. This includes co-Principal Investigator (PI) Cheryl Nickerson and her team Jennifer Barrila (co-I) and Jiseon Yang (originally a postdoc, now Research Assistant Professor) at Arizona State University (ASU), as well as C Mark Ott (co-I) at NASA Johnson Space Center (NASA-JSC). Two additional participants, Sandhya Gangaraju and Richard Davis at Arizona State University (ASU) are also participating. Due to the expertise of our team, we performed all the necessary microscopy and will not require external collaborators as originally considered. In the PI's lab at Texas State (TxST), Starla Thornhill was a master's degree student as the grant was being written. She successfully defended her MS thesis in 2016 and completed her PhD in 2020 on work directly related to this grant. She continued as a postdoc during 2021 and along with Sandhya Gangaraju and Jiseon Yang (from ASU), completed all the postflight microscopy. As well, we have added Alistair McLean as an unpaid collaborator. He works as a software engineer and wrote a program that helps us enumerate bacterial populations from microscopy images. Finally, as a result of discussions during the flight readiness review, we have been asked to consult with metallurgical experts from NASA Marshall Space Flight Center (NASA-MSFC) to assist our team in interpreting microbial corrosion information. We agree to this request as it will greatly enhance the science and benefits to NASA and our team. The specific metallurgical expert(s) are being identified by S. Gorti at NASA-MSFC.		

COI Name (Institution):	Barrila, Jennifer Ph.D. (Arizona State University) Ott, C. Mark Ph.D. (NASA Johnson Space Center) Nickerson, Cheryl Ph.D. (Arizona State University)
Grant/Contract No.:	NNX17AC79G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Bacteria and other microorganisms have been shown to grow as surface-adherent biofilm communities in the microgravity environment during spaceflight. Organisms within biofilms are quite resistant to traditional antimicrobial chemicals and can foul water treatment filters resulting in potential risk to the astronauts. The biofilms on International Space Station (ISS) consist of many species, yet studies to date have only tested single-species biofilms. There is growing evidence that mixed-culture biofilms behave quite differently from pure culture biofilms, and we anticipate this phenomenon to occur in the microgravity environment of ISS. Through this study we will investigate the structure by which two-species, mixed-culture biofilms form under microgravity on stainless steel used to construct Environmental Controls and Life Support Systems (ECLSS) on the ISS. As well, we will explore the efficacy by which mixed-culture biofilms can be controlled, and whether microgravity-grown mixed-culture biofilms cause corrosion on stainless steel. As a result of supplemental funding, we are now able to gather genetic data on the mechanisms by which mixed bacterial species are able to form biofilms in microgravity conditions. Our results are designed to both enable future spaceflight missions and translate into information for the general scientific community on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	October 2021 Report: Our spaceflight experiment launched on SpaceX CRS-21 (SpX-21) on December 6, 2020 and two of our three samples returned on January 14, 2021. The third sample returned with Crew-1 on May 2, 2021. We encountered two major issues that have led us to request a reflight (that decision is currently pending). One issue was media contamination, rendering our data interpretation as inconclusive. A second issue (needlestick injury to a space crewmember) resulted in some of our samples incubating for 117 days on the International Space Station (ISS) rather than the planned 4 days. While that caused a potential loss of some transcriptome data (described in the Task Progress section), a really positive development from the extended incubation resulted in enhanced microbial corrosion data, as well as the return of viable microorganisms that can now be investigated for genetic changes as a consequence of prolonged spaceflight. While data analysis will continue, our preliminary results suggest that corrosion might be enhanced during spaceflight.
Task Progress:	Our flight experiment was authorized in June 2020 and our experiment, entitled “Bacterial Adhesion and Corrosion” (BAC) launched to the ISS on Space X-21 (CRS-21) on 12/6/2020. Our objectives were to: examine and compare mixed culture bacterial biofilm growth on Earth and during spaceflight; investigate susceptibility of biofilms to a proposed silver-based disinfectant; measure corrosion due to biofilm growth; and measure gene expression that would indicate bacterial responses to the various growth conditions (microgravity on the ISS, full gravity on Earth, and the presence or absence of silver disinfectant). Experimental materials for spaceflight and ground control studies were prepared at Kennedy Space Center (KSC) prior to launch. The ground control experiments were conducted using the same protocols as were done in space. Our original plan was to sample three time points (4 days, 14 days, and 24 days). All three samples were incubated on the ISS as scheduled, but due to crew time limitations only two of the samples (14-day and 24-day) were processed and returned with CRS-21 on 1/14/2021. The third sample (originally planned for a 4-day incubation) returned with the Crew-1 payload on 5/2/2021 (total incubation 117-days). After the experiment was loaded onto the rocket and just prior to launch, we were informed that one component of our growth media components (an amino acid solution) had failed the manufacturer’s sterility test. When we analyzed our ground control samples and compared those to previous samples incubated under identical conditions, there were significant differences. Further testing with media prepared with the contaminated amino acid showed a different bacterial growth profile when compared to bacterial growth in uncontaminated media. Analysis of biofilm colonization images from all three time points (14-day and 24-day, returned on CRX-21; and 117-day, returned on Crew-1) showed no significant difference between spaceflight and ground control samples in the absence of silver disinfection. However, differences were observed between spaceflight and ground control biofilm samples grown in the presence of silver disinfectant. However, given the presence of an unknown media contaminate we cannot attribute whether the biofilm images seen are due to spaceflight or attributable to media contamination. The unexpected change in our third sample (originally planned for 4-days, returned after 117-days) resulted in some changes to our available data. While we did not get the initial high yield of RNA (anticipated with a 4-day culture), our cultures were returned without fixation and as a result a number of viable organisms were obtained. These cultures will represent a unique opportunity to investigate the role of long-term culture (i.e., 117-days) on microgravity-grown, when compared to ground-controls (i.e., grown in full gravity). We also have the original ancestral bacteria available and plan on sequencing these organisms in the near future to take advantage of this unique opportunity. Examination of microbial corrosion of the stainless steel used in our experiments showed enhanced results that would not have been available in shorter-term trials. Statistical analysis of the corrosion data is ongoing and initial impressions (subject to validation) imply an enhancement of corrosion in microgravity. In summary, while we obtained some very interesting results, the media contamination is a real concern, so we are unable to accurately interpret the data from the Bacterial Adhesion and Corrosion (BAC) mission. Nevertheless, our unexpected long-term (117-day) sample provides some unexpected positive benefits with enhanced corrosion results and also the return of viable long-term spaceflight grown cultures that can be sequenced to identify genetic changes that might occur due to spaceflight. As stated earlier, we have requested a reflight due to the media contamination concerns (and have adopted contingency plans to avoid this). In the event that a reflight is not approved, we will request a no-cost extension so that we can complete our genetic (transcriptome) analysis, since those funds were just released.
Bibliography Type:	Description: (Last Updated: 11/20/2025)

Articles in Peer-reviewed Journals	Yang J, Barrila J, Ott CM, King O, Bruce R, McLean RJC, Nickerson CA. "Longitudinal characterization of multispecies microbial populations recovered from spaceflight potable water." npj Biofilms and Microbiomes. 2021 Sep 6;7:70. https://doi.org/10.1038/s41522-021-00240-5 ; PubMed PMID: 34489467; PMCID: PMC8421509 , Sep-2021
Articles in Peer-reviewed Journals	McLean RJC, Brown ET. "Potential influences of bacterial cell surfaces and nano-sized cell fragments on struvite biomineralization." Crystals. 2020 Aug 15;10(8):706. https://doi.org/10.3390/cryst10080706 , Aug-2020
Articles in Peer-reviewed Journals	Nickerson CA, Medina-Colorado AA, Barrila J, Poste G, Ott CM. "A vision for spaceflight microbiology to enable human health and habitat sustainability." Nat Microbiol. 2022 Apr;7:471-4. https://doi.org/10.1038/s41564-021-01015-6 ; PMID: 34903836 , Apr-2022
Dissertations and Theses	Thornhill SG. (Starla G. Thornhill). "Designing and Optimizing a Biofilm Experiment for Space Flight." PhD Thesis, Texas State University, November 2020. , Nov-2020